UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

9476785025

BIOLOGY **5090/02**

Paper 2 Theory **May/June 2008**

1 hour 45 minutes

Candidates answer Section A on the Question Paper.
Additional Materials: Answer Booklet/Paper

READ THESE INSTRUCTIONS FIRST

Write your Centre number, candidate number and name on all the work you hand in.
Write in dark blue or black pen.
You may use a pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.
DO **NOT** WRITE IN ANY BARCODES.

Section A
Answer **all** questions.
Write your answers in the spaces provided on the Question Paper.
Section B
Answer **all** the questions including questions 6, 7 and 8 **Either** or 8 **Or**.
Write your answers on the separate Answer Paper provided.
Write an **E** (for Either) or an **O** (for Or) next to the number 8 in the examiner's grid below to indicate which question you have answered.

You are advised to spend no longer than one hour on Section A and no longer than 45 minutes on Section B.
At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
Section A	
Section B	
6	
7	
8	
Total	

This document consists of **11** printed pages and **1** blank page.

© UCLES 2008

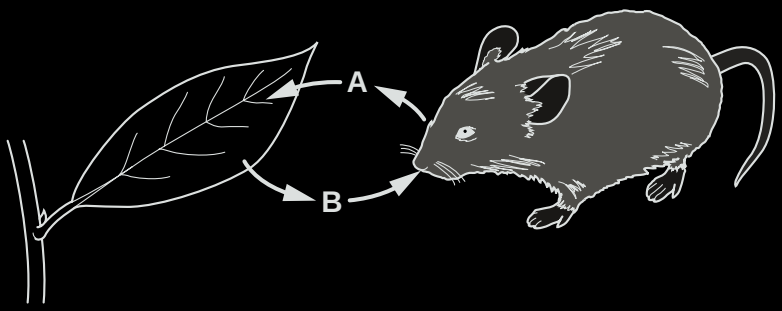


Fig. 1.1a

(a) Name

(i) gas **A**, ...

(ii) gas **B**. ..... [2]

(iii) On Fig. 1.1b, place arrow heads on the **four** lines to indicate the movement of the same gases during the hours of darkness. [3]

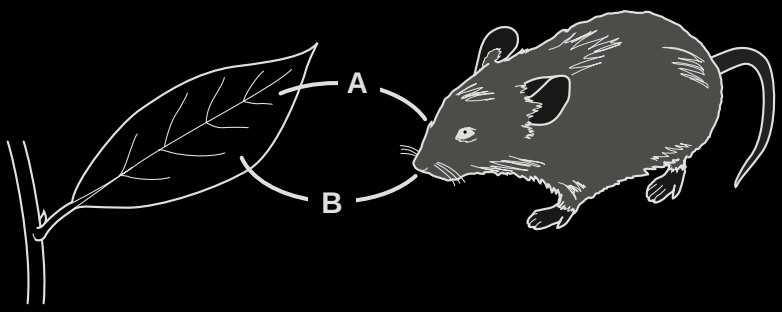


Fig. 1.1b

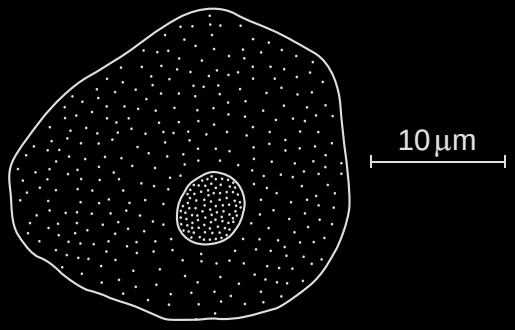


Fig. 2.1

(a) Name the chemical found in the nucleus that controls the production of protein.

..... [1]

Fig. 2.2 shows a gamete from the same person.

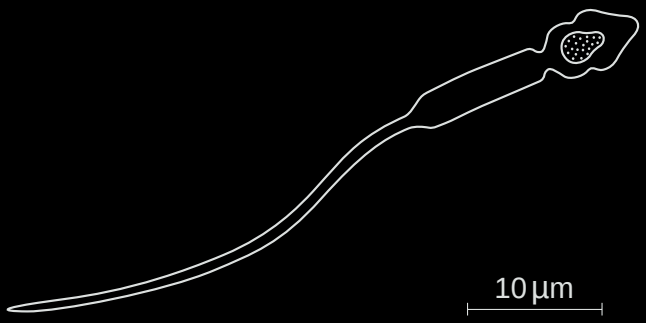


Fig. 2.2

(b) Describe how and explain why the two cells differ in appearance.

.....

.....

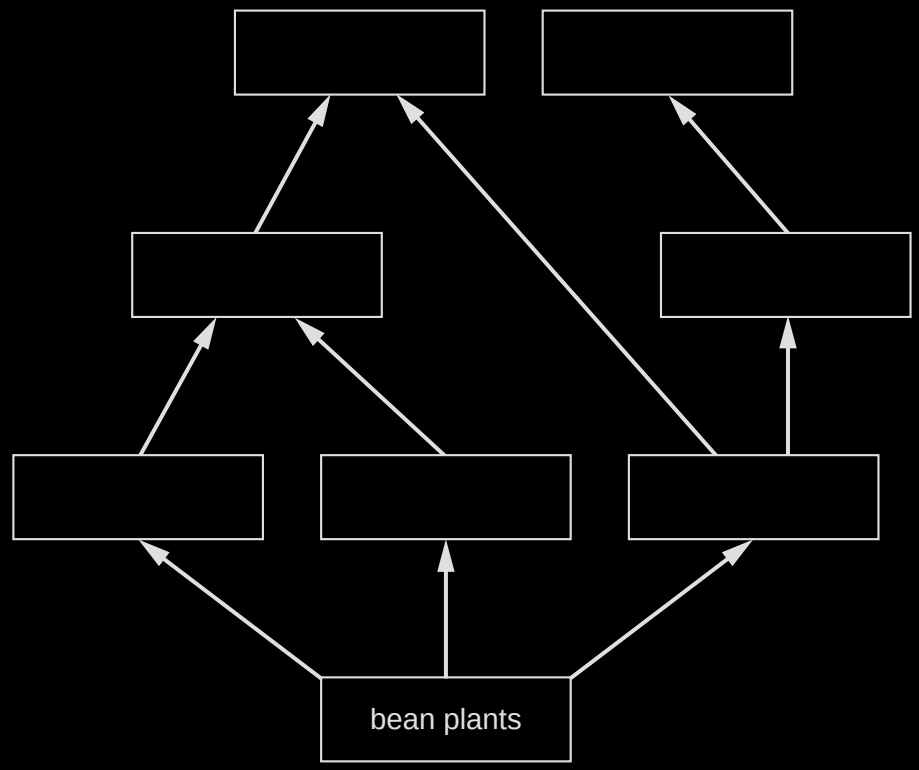
.....

.....

..... [4]

1. *Young shoots of a crop of bean plants covered with greenflies (aphids) sucking food from the stems.*
2. *Saw a large bird (hawk), which usually catches mice, swoop to take a small yellow bird clinging to a bean stem. Noticed that these small birds often visit the bean field to eat some of the aphids or butterflies.*
3. *Flowers of beans being visited by many different species of butterfly.*
4. *Mice seen nibbling at some dispersed bean seeds.*
5. *Spider's web constructed between two bean plants with 5 large black flies caught in it. Rotting body of a mouse nearby attracting similar flies.*

(a) Complete Fig. 3.1 by filling in the names of the organisms to show the feeding relationships in this community.



[4]

Fig. 3.1

(b) **(i)** What name is given to a chart of feeding relationships as shown in Fig. 3.1?

.....

[1]

(ii) Name two top carnivores observed by the student.

1. .....

2. .....

[2]

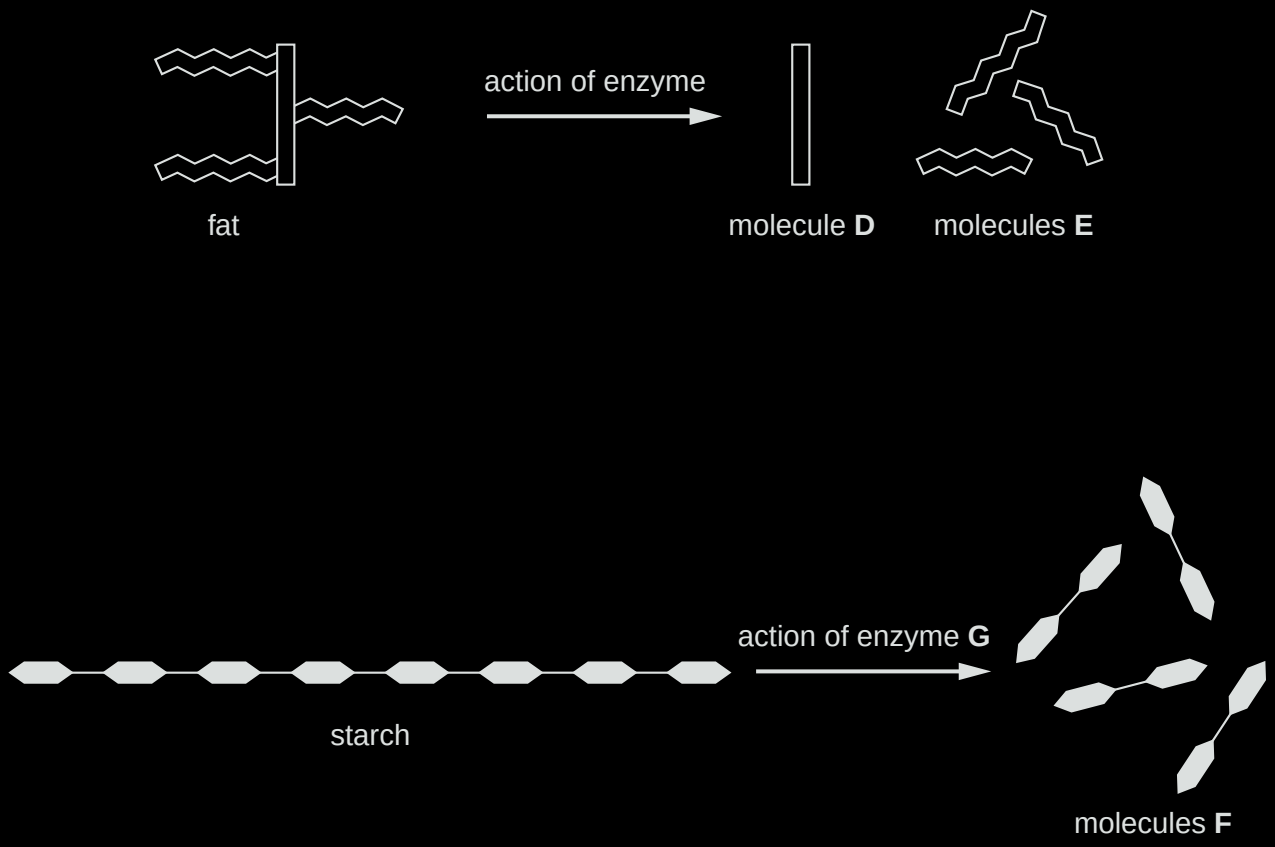


Fig. 4.1

(a) Identify the molecules shown in Fig. 4.1.

D .....

E .....

F ..... [3]

(b) Identify enzyme **G**. ..... [1]

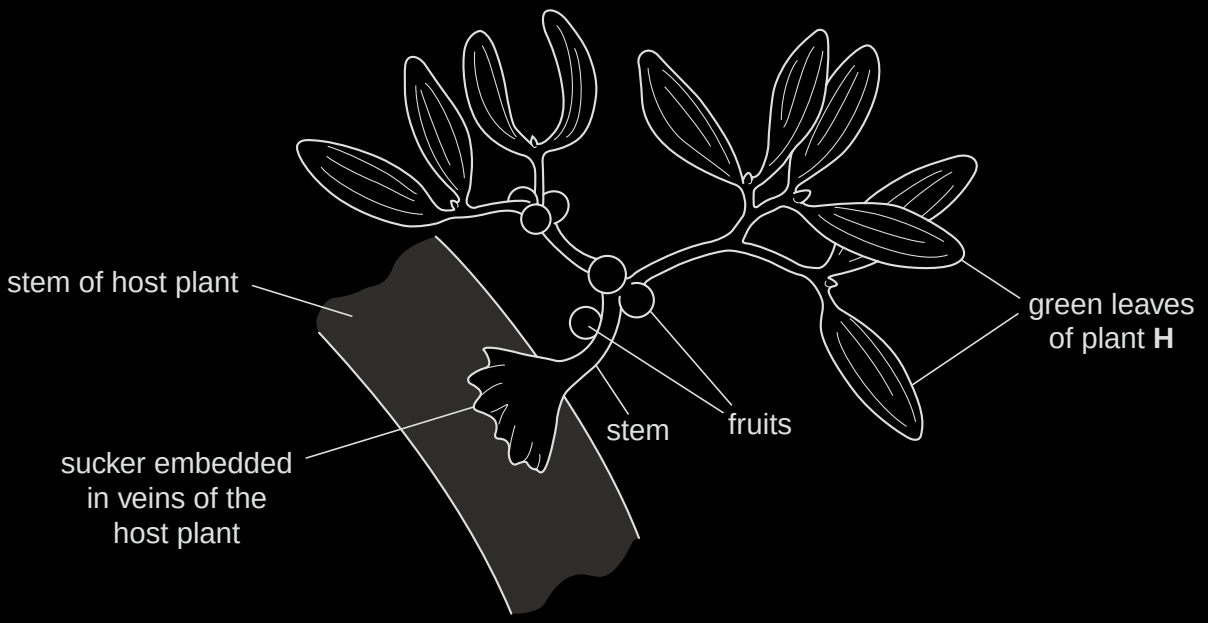


Fig. 5.1

(a) (i) Name an organ, found in free-living plants, that is absent in plant **H**.

..... [1]

(ii) Name the process used by plant **H** to manufacture its carbohydrates.

..... [1]

Plant **H** obtains one raw material for food manufacture from the air and other requirements, as well as nutrients, from its host.

(b) (i) Suggest three substances it absorbs from its host, and for each, state its use in Plant **H**.

substance 1 ....., use .....

substance 2 ....., use .....

substance 3 ....., use .....
[6]

(ii) Suggest the name of the tissue in the host plant which supplies these substances to plant **H**.

..... [1]

[Total: 9]